

Instrument Transformers

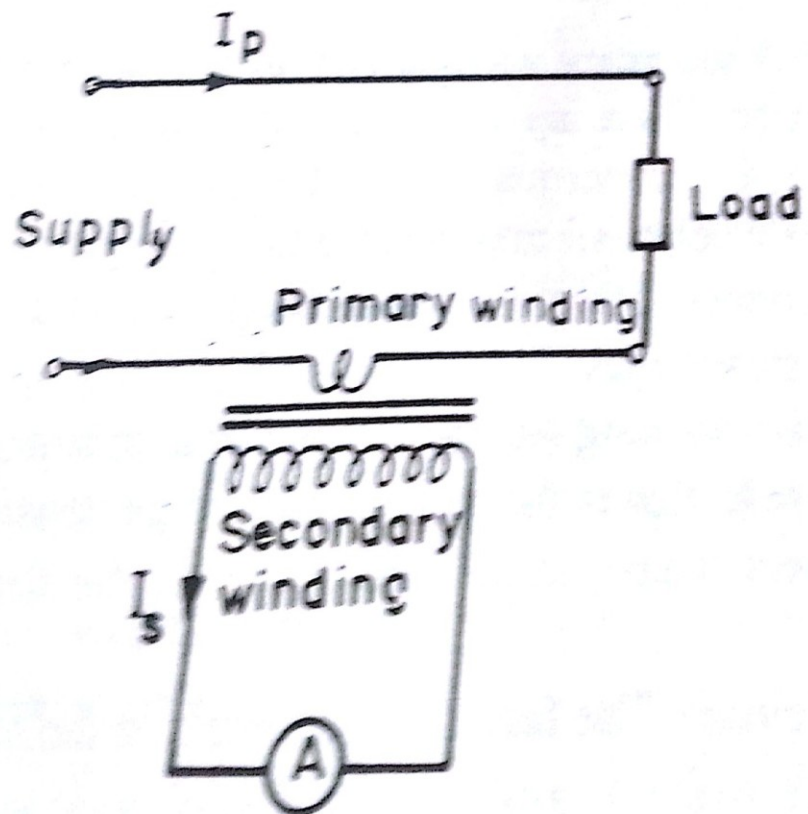
Definition

- Transformers used in conjunction with measuring instruments for measurement purposes.

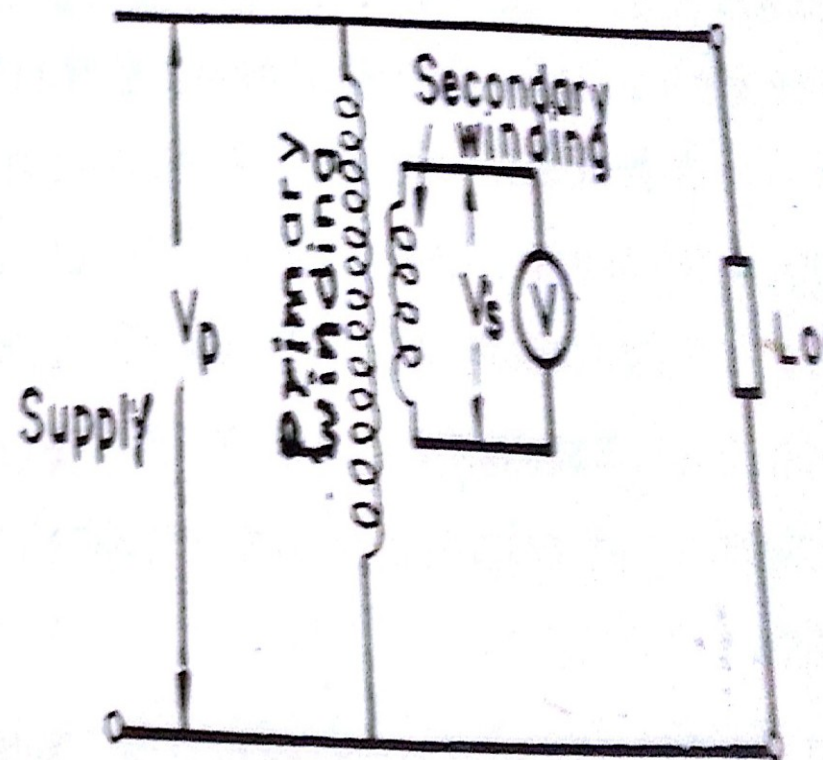
Uses

- Used for range extension because:
 - * Instruments of moderate size can be used.
 - * Meters can be standardized.
 - * Metering circuit is isolated from high voltage circuit, thereby assuring safety.

CT



PT

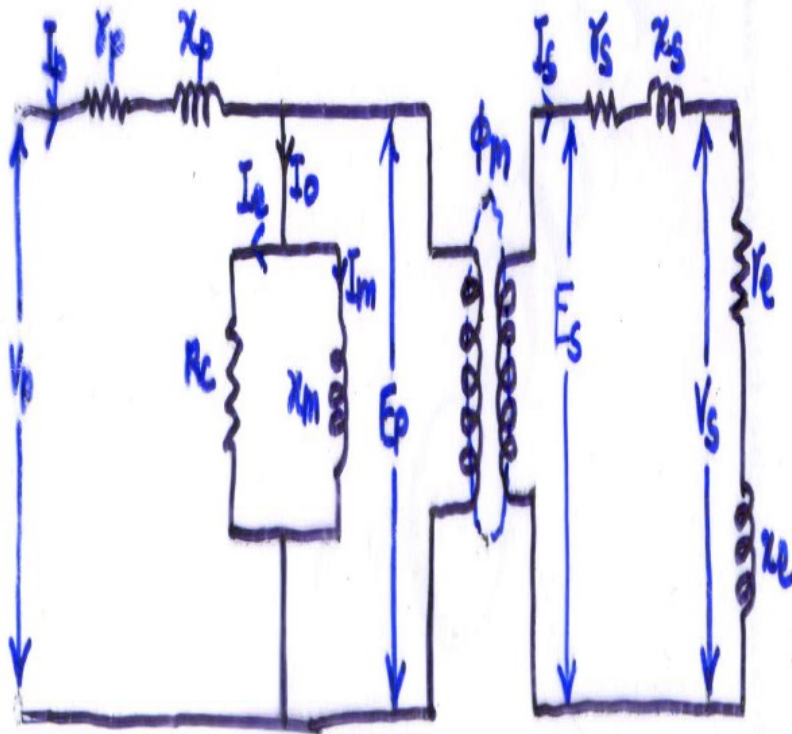


Difference between CT and PT

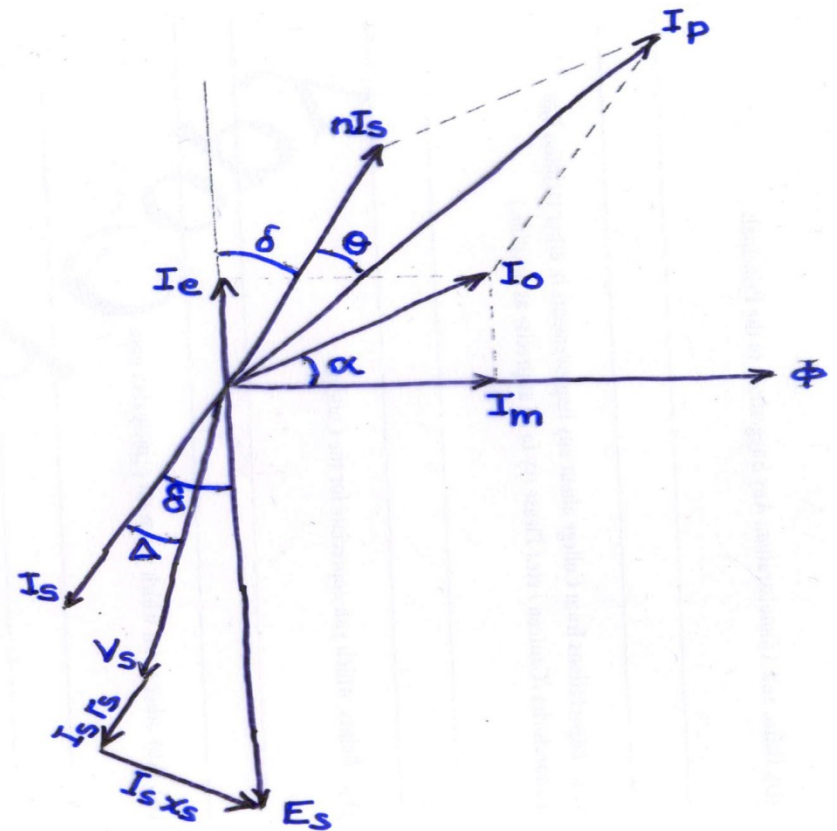
- I_p in a CT is independent of the secondary winding circuit conditions.
- I_p in a PT depends upon secondary circuit burden.

CT

Equivalent Circuit



Phasor Diagram



$n \rightarrow$ turns ratio

$\theta \rightarrow$ phase angle of transformer

$\phi \rightarrow$ phase angle of total burden

$$= \tan^{-1} \left(\frac{x_s + x_e}{r_s + r_e} \right)$$

$\Delta \rightarrow$ phase angle of external burden $= \tan^{-1} \frac{x_e}{r_e}$

Ratios

$$\begin{aligned}\text{Transformation Ratio (R)} &= \frac{|\text{Primary Phasor}|}{|\text{Secondary Phasor}|} \\ &= \frac{\text{Primary Winding Current}}{\text{Secondary Winding Current}} \quad \text{for C.T.} \\ &= \frac{\text{Primary Winding Voltage}}{\text{Secondary Winding Voltage}} \quad \text{for P.T.}\end{aligned}$$

$$\text{Nominal Ratio } (K_n) = \frac{\text{Rated Primary Winding Current}}{\text{Rated Secondary Winding Current}} \quad (\text{for C.T.})$$

$$= \frac{\text{Rated Primary Winding Voltage (for P.T.)}}{\text{Rated Secondary Winding Voltage}}$$

$$\text{Ratio Correction Factor (RCF)} = R/K_n$$

BURDEN OF AN INSTRUMENT TRANSFORMER

$$\text{Total Secondary winding burden} = \frac{(\text{Secondary winding induced voltage})^2}{(\text{impedance of secondary winding circuit including impedance of secondary winding})}$$

$$= (\text{secondary winding current})^2 \times (\text{impedance of secondary winding circuit including secondary winding})$$

$$\text{Secondary winding burden due to load} = \frac{(\text{Secondary Winding Terminal Voltage})^2}{(\text{impedance of load on secondary winding})}$$

$$= (\text{Secondary winding current})^2 \times (\text{impedance of load in the secondary winding circuit})$$

Relationships in CT

$$R = \frac{I_p}{I_s} \approx n + \frac{I_o}{I_s} \sin(\delta + \alpha) \approx n + \frac{I_m \sin \delta + I_e \cos \delta}{I_s}$$

$$\text{Phase Angle } (\theta) \approx \frac{I_o \cos(\delta + \alpha)}{n I_s} \text{ rad} \approx \frac{I_m \cos \delta - I_e \sin \delta}{n I_s} \text{ rad}$$

Errors in CT

(i) Ratio Error

$$\text{Ratio Error} = \frac{K_n - R}{R} \approx n + \frac{I_e}{I_s} \approx n \left(1 + \frac{I_e}{I_p} \right)$$

$(\because I_p \approx n I_s)$

(ii) Phase Angle Error

$$\theta \approx \frac{I_m}{n I_s} \approx \frac{I_m}{I_p} \text{ rad}$$

Causes of Errors in CTs

- Exciting mmf is required by the primary winding to produce flux. Hence, transformer draws a magnetizing current.
- Core losses and copper losses
- Magnetic leakage

Important!

Turns Ratio

$$n = \frac{N_2}{N_1} = \frac{I_1'}{I_s} \quad (\text{For C.T.}) = \frac{N_s}{N_p}$$

$$n = \frac{N_1}{N_2} = \frac{I_s}{I_1'} \quad (\text{For P.T.}) = \frac{N_p}{N_s}$$